

ORGANIC REAGENTS

S.No.	Reagent	Function
1	PCl ₃ , PBr ₃ , PI ₃	Alcohols into Alkyl halides
2	SOCl ₂ , PCl ₅	Alcohols into Alkyl chlorides & Carboxylic acids into Acid Chlorides
3	HCl/ZnCl ₂ , HBr, HI	Alcohols into alkyl halides
4	Cl ₂ /Fe or FeCl ₃	Cl group substituting on Benzene
5	NaNO ₂ /HCl 0-5 ⁰ C	Diazotisation
6	CuCl, CuBr, CuCN, KI, H ₂ O, H ₃ PO ₂	Diazonium Chloride into Chlor Benzene, Bromo Benzene, Benzo nitrile, Iodo Benzene, Phenol, Benzene respectively
7	HBF ₄ or NaBF ₄	Diazonium Chloride into Fluoro Benzene
8	AgF or Hg ₂ F ₂ or SbF ₃ or CoF ₂	Alkyl halides into alkyl fluorides
9	Na / dry ether	Alkyl halides into alkanes
10	NaOH 623/443/368K	Chloro benzene to phenol
11	Br ₂ /FeBr ₃	Bromination of Benzene
12	Cl ₂ /FeCl ₃	Chlorination of Benzene
13	CH ₃ Cl /AlCl ₃	alkylation of benzene and its derivatives
14	CH ₃ -CO-Cl /AlCl ₃	Acylation on benzene
15	H ₂ SO ₄ /HNO ₃	Nitration of benzene
16	(CH ₃ CO) ₂ O /AlCl ₃	O ⁻ Acylation of Phenol
17	H ₂ SO ₄	Sulphonation on Benzene
18	H ₂ O/ H ₂ SO ₄	alkenes into alcohols
	Aq KOH	Alkyl halide into alcohol
19	BH ₃ /H ₂ O ₂ /OH ⁻	Alkenes into alcohols (Anti Markownikoff product)
20	NaBH ₄ / LiAlH ₄ (LAH)	Aldehydes, ketones, acids into alcohols, Nitriles & Cyanides, Isocyanides into amines
21	H ₂ / Ni or H ₂ /Pd	reduction of aldehydes, ketones and cyanides
22	RMgX/H ₃ O ⁺	Aldehydes, ketones into alcohols
23	O ₂ /H ⁺	Cumene to phenol
24	Na	Alcohol or phenol into Sodium alkoxide/Phenoxide
25	(CH ₃ CO) ₂ O/ CH ₃ -CO-Cl	O acylation on phenol or N acylation on Aniline or amine
26	Conc.H ₂ SO ₄ /443K	Conversion of primary alcohols into Alkenes
27	Conc.H ₂ SO ₄ /410K	Conversion of alcohols into Ethers
28	85% H ₃ PO ₄ / 440K	Secondary alcohol into alkene
29	20% H ₃ PO ₄ /358K	Tertiary alcohol into alkene
	Alcoholic KOH	Alkyl halide into alkene

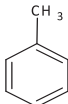
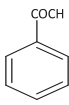
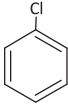
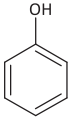
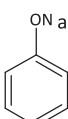
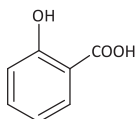
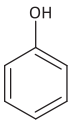
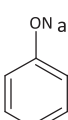
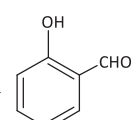
ORGANIC REAGENTS

30	CrO ₃ /KMnO ₄ or K ₂ Cr ₂ O ₇ in acidic medium	oxidation of alcohols into acids
31	Cu /573K	dehydrogenation of alcohols gives 1 ⁰ alcohols into aldehydes and 2 ⁰ alcohols into ketones & 3 ⁰ alcohols into alkenes
32	Dil. HNO ₃	Mono nitration of Phenol
33	Conc.HNO ₃	tri nitration of phenol
34	Br ₂ /H ₂ O	tri bromination of phenol
35	Br ₂ /CS ₂	mono bromination of phenol
36	NaOH /CO ₂	Phenol to salicylic acid
37	CHCl ₃ /NaOH	Phenol to salicylaldehyde
38	Zn dust	Phenol to Benzene
39	Na ₂ Cr ₂ O ₇ /H ₂ SO ₄ or air	Phenol to Benzo quinone
40	Zn/Cr ₂ O ₃ 200 to 300 atm 573 ~ 673K	CO & H into methanol
41	Invertase	Sucrose into Glucose or Fructose
42	Zymase	Glucose or Fructose into ethanol
43	HI	Ether into alcohol & alkyl halide
44	PCC	alcohol to aldehyde
45	Pd /BaSO ₄ , H ₂	acid chloride into aldehyde
46	SnCl ₂ /HCl/H ₃ O ⁺	Cyanides into aldehydes
47	AlH(i-Bu) ₂ /H ₂ O	Cyanides into aldehydes
48	DIBAL-H/H ₂ O	Esters into aldehydes
49	CrO ₂ Cl ₂ /H ₂ O	Toluene to aldehyde
50	CrO ₃ / (CH ₃ CO) ₂ O	Toluene into Benzaldehyde
51	Cl ₂ /hv	Chlorination on alkyl group of Benzene or alkane
52	CO, HCl anhydrous AlCl ₃	Benzene to Benzaldehyde
53	(CH ₃) ₂ Cd	acid chloride into ketones
54	RMgX/H ₃ O ⁺	Cyanides into ketones
55	HCN	Carbonyl compound into cyanohydrin
56	NaHSO ₃	addition to aldehyde and ketone
57	H ₂ NOH	carbonyl compound into oxime
58	H ₂ N-NH ₂	carbonyl compound into hydrazone
59	H ₂ N-NH-Ph	carbonyl compound into Phenyl hydrazone
60	2,4DNP	carbonyl compound into 2,4 dinitro phenyl hydrazone
61	H ₂ N-NH-CO-CH ₃	carbonyl compound into semi carbazide
62	ROH/HCl	Aldehydes & ketones into hemiacetal and acetal
63	HO-CH ₂ -CH ₂ -OH/HCl	Aldehyde or ketone into ethylene glycol

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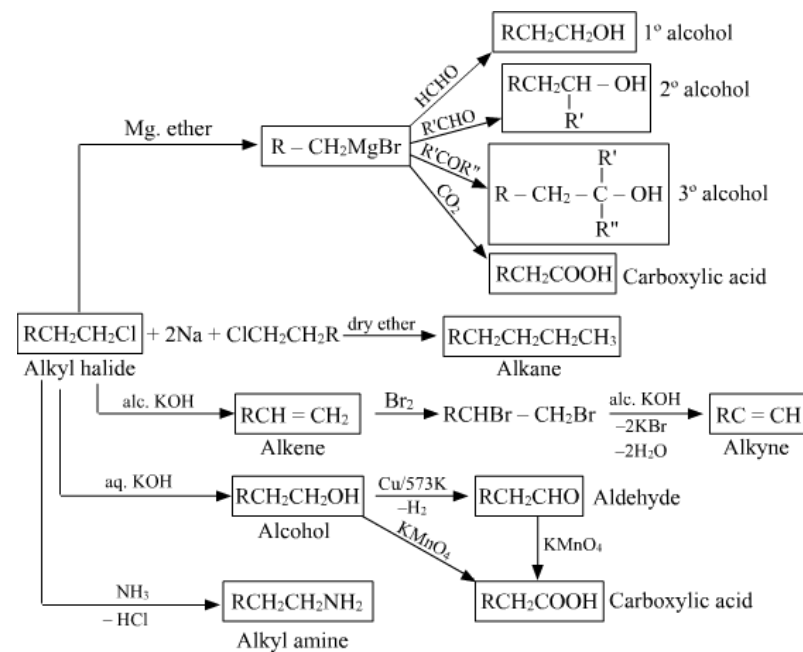
		ketone
64	Zn-Hg/HCl	carbonyl compound into alkane
65	H ₂ N-NH ₂ /KOH	carbonyl compound into alkane
66	KMnO ₄ /OH ⁻ /K ₂ Cr ₂ O ₇ /H ₂ SO ₄ or HNO ₃	Ketones into mixture of carboxylic acids on prolonged oxidation
67	(Ag(NH ₃) ₂)NO ₃ +NaOH	Tollen's test
68	Cu(OH) ₂	Fehling's test
69	NaOH+I ₂	Iodoform
70	NaOH or Ba(OH) ₂	aldal condensation
71	Conc KOH or NaOH	Cannizaro's reaction
72	KMnO ₄ /KOH	Toluene/alkyl Benzene into Benzoic Acid
73	H ₂ O/H ⁺	Cyanides into carboxylic acids, amides into carboxylic acids, esters into carboxylic acid and alcohols, acid chlorides or anhydrides into carboxylic acids
74	NaOH	Saponification of ester, acid into salt of acid
75	Na ₂ CO ₃ or NaHCO ₃	Carboxylic acid test
76	P ₄ O ₁₀ or P ₂ O ₅	Dehydration of acids into anhydride, amides into nitriles
77	ROH/conc H ₂ SO ₄	Carboxylic acids into esters
78	PCl ₃ , SOCl ₂ , PCl ₅	Carboxylic acids into acid chlorides
79	NH ₃ heating	Carboxylic acids into amides
80	NaOH/CaO	Decarboxylation (acids into alkanes)
81	LiAlH ₄	Carboxylic acids into alcohols, amides into amines
82	Cl ₂ /red.P ₄	HVZ reaction
83	Sn/HCl or Fe/HCl, H ₂ /Pd	Reduction of nitro compounds into amines
84	NH ₃	Alkyl halides into amines
85	H ₂ /Ni or H ₂ /Pd LiAlH ₄	Amides into cyanides
86	KOH/R-X	Phthalamide into amine
87	NaOH/Br ₂	Hoffman bromamide, amide into amine with one :C ⁻ less
88	KOH,CHCl ₃	Amines into Carbyl amines
89	NaNO ₂ /HCl	1 ^o aliphatic amines into alcohols
90	NaNO ₂ /HCl 0 - 5 ^o C	Aniline into diazonium chloride
91	C ₆ H ₅ SO ₂ Cl	Distinguishing 1 ^o ,2 ^o & 3 ^o amines
92	Br ₂ /H ₂ O	Aniline into tri bromo aniline
93	Br ₂ /CH ₃ -CO-Cl/(CHCO) ₂ O	Aniline into Bromo Aniline

NAME REACTIONS

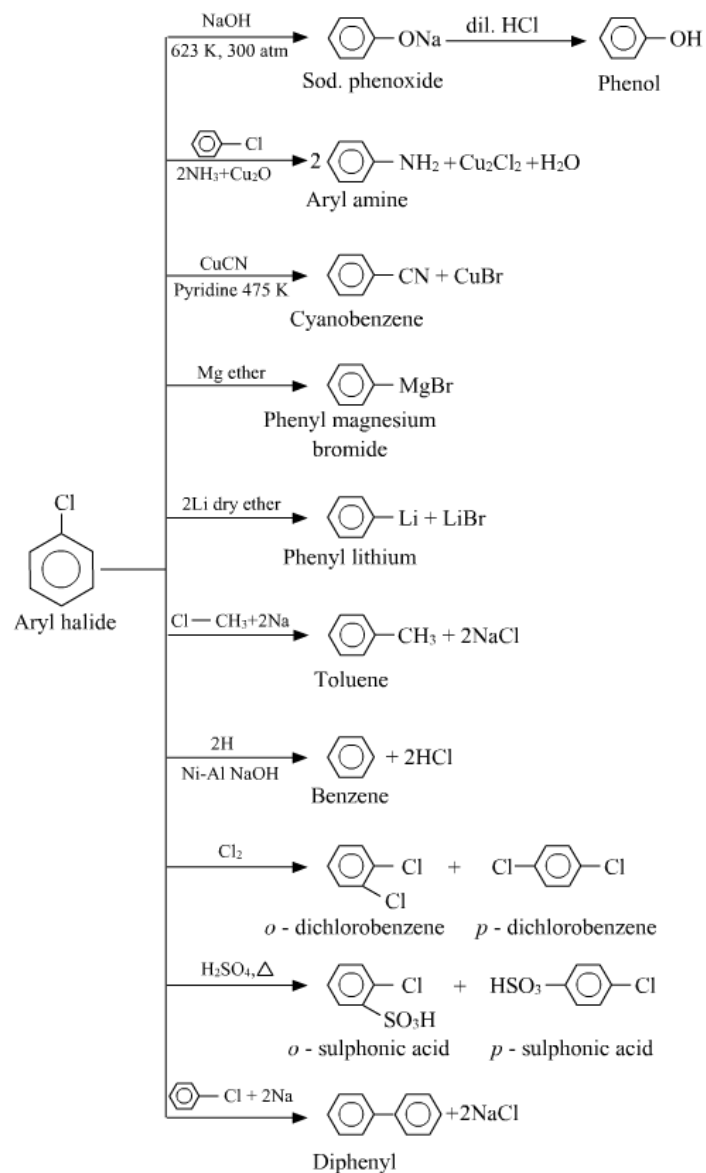
1.	Finkelstein	$\text{CH}_3\text{Br} + \text{NaI} \xrightarrow{\text{Acetone}} \text{CH}_3\text{I} + \text{NaBr}$
2.	Swarts	$\text{CH}_3\text{Br} + \text{AgF} \longrightarrow \text{CH}_3\text{F} + \text{AgBr}$
3.	Friedel-Crafts Alkylation	 + $\text{H}_3\text{C}-\text{Cl}$ $\xrightarrow{\text{Anhydrous AlCl}_3}$ 
4.	Friedel-Crafts Acylation	 $\xrightarrow[\text{Anhydrous AlCl}_3]{\text{CH}_3\text{COCl}}$ 
5.	Wurtz	$\text{H}_3\text{C}-\text{Cl} + \text{Cl}-\text{CH}_3 \xrightarrow[\text{Dry ether}]{2\text{Na}} \text{H}_3\text{C}-\text{CH}_3 + \text{NaCl}$
6.	Fittig	 +  $\xrightarrow[\text{Dry ether}]{2\text{Na}} \text{Ph}-\text{Ph} + \text{NaCl}$
7.	Wurtz-Fittig	 + $\text{Cl}-\text{CH}_3 \xrightarrow[\text{Dry ether}]{2\text{Na}} \text{Ph}-\text{CH}_3 + \text{NaCl}$
8.	Kolbe	 $\xrightarrow{\text{NaOH}}$  $\xrightarrow[\text{ii) H}^+]{\text{i) CO}_2}$ 
9.	Reimer-Tiemann	 $\xrightarrow{\text{CH}_3\text{Cl} + \text{NaOH}}$  $\xrightarrow{\text{H}^+}$ 
10.	Williamson	$\text{CH}_3-\text{Br} + \text{CH}_3-\text{ONa} \longrightarrow \text{CH}_3-\text{O}-\text{CH}_3 + \text{NaBr}$
11.	Stephen	$\text{H}_3\text{C}-\text{CN} + \text{SnCl}_2 + \text{HCl} \longrightarrow \text{H}_3\text{C}-\text{CH}=\text{NH} \xrightarrow{\text{H}_3\text{O}^+} \text{H}_3\text{C}-\text{CHO}$

7. All primary amines ($R/Ar-NH_2$) give **Carbyl Amine Test**
 $R-NH_2 + CHCl_3 + KOH(alc) \rightarrow R-NC + KCl + H_2O$
 offensive smell
8. Aniline gives Azo Dye Test (Only for aromatic amines)
 $C_6H_5NH_2 + NaNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^-$; then add β -naphthol orange dye
9. All alcohols (**ROH**) give **Na-metal test**
 $R-OH + Na \rightarrow R-ONa + H_2$
 bubbles
10. For esters ($RCOOR$) : Hydrolyses first. Then see the products (acid & alcohol) and give a test to identify them.
11. All alkenes ($C=C$) and alkynes ($C\equiv C$) decolorizes Br_2 – water from red to colourless
12. **Lucas Test to distinguish primary, secondary and tertiary alcohols**
 Lucas reagent: $ZnCl_2/HCl$
 3° -alcohol + Lucas reagent $\rightarrow$ immediate turbidity
 2° -alcohol + Lucas reagent $\rightarrow$ turbidity after sometime
 1° -alcohol + Lucas reagent $\rightarrow$ no turbidity

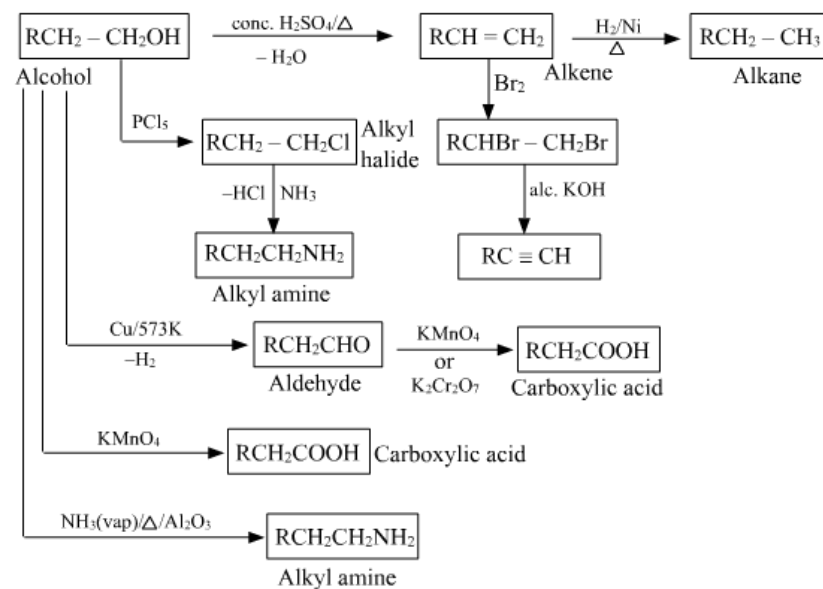
SCHEME – I: Conversions related to alkyl halides



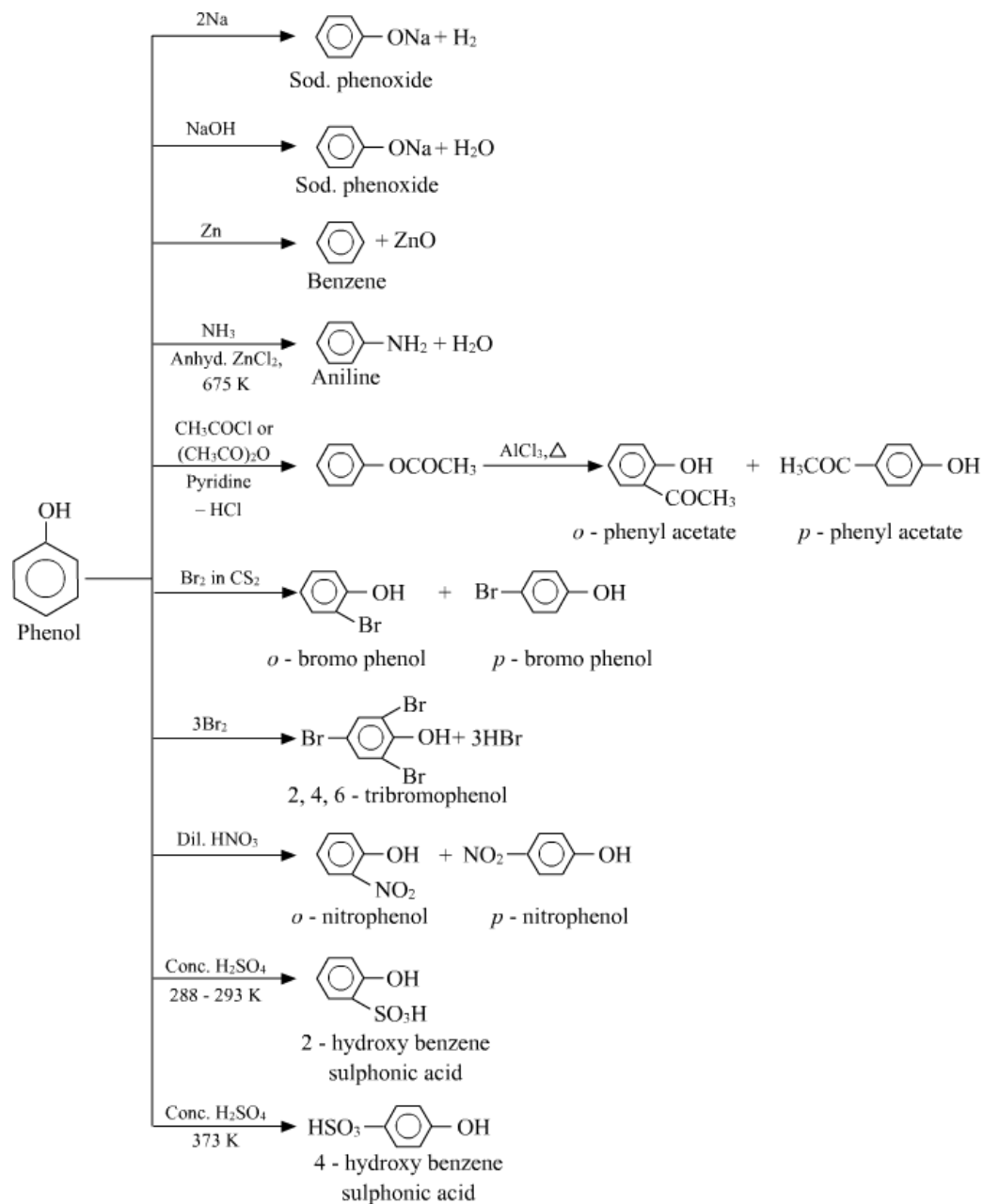
SCHEME – II: Conversions related to aryl halides



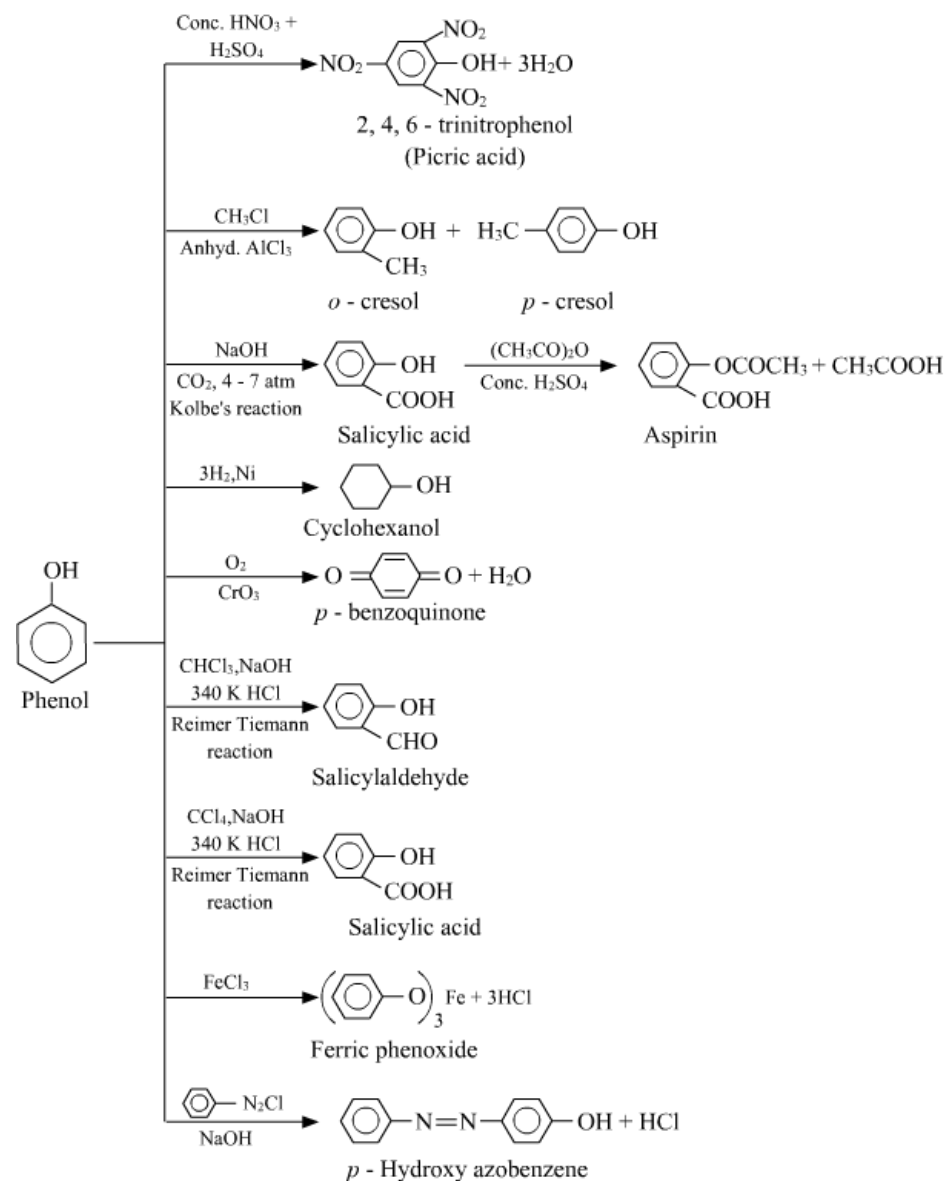
SCHEME – III: Conversions related to alcohols



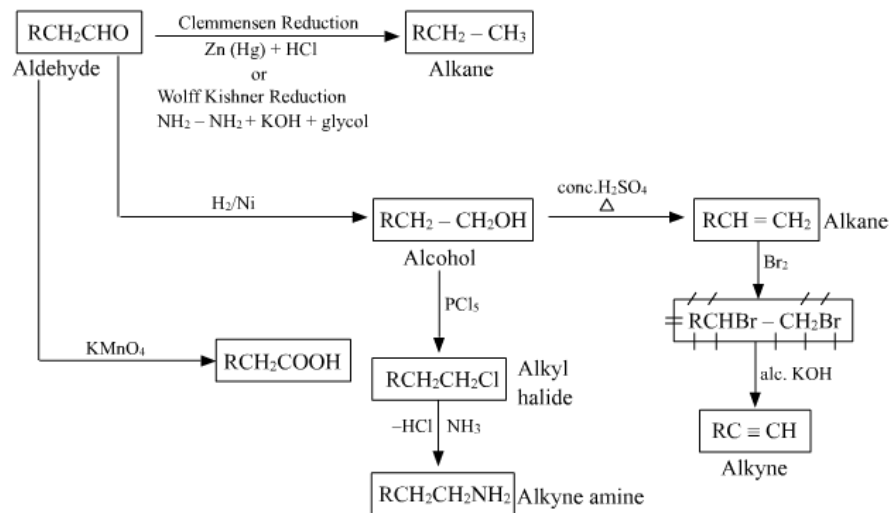
SCHEME – IV: Conversion related to phenols – I



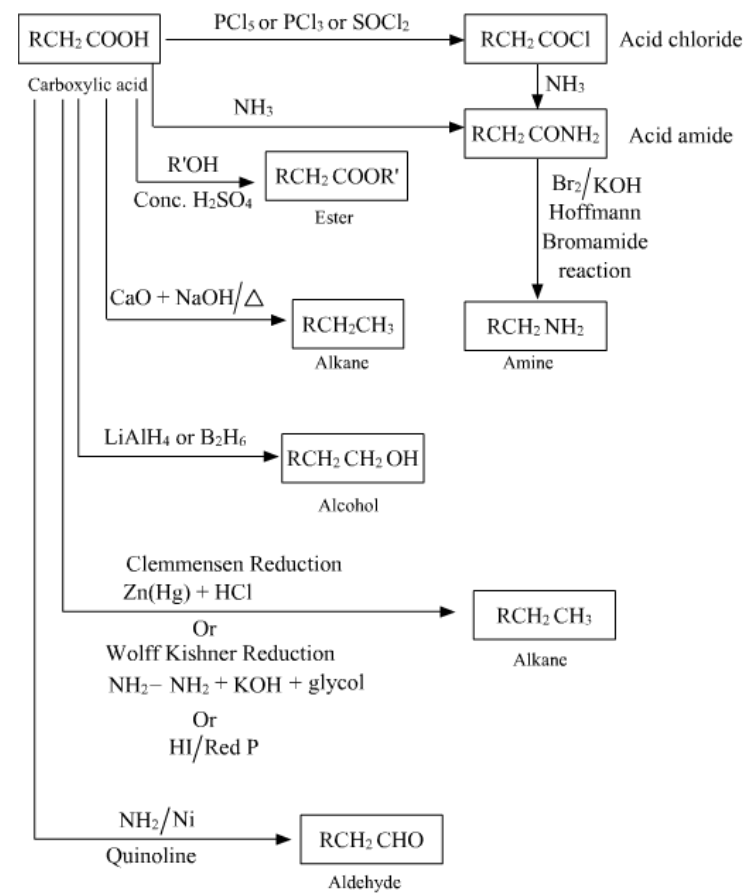
SCHEME – V: Conversion related to phenols – II



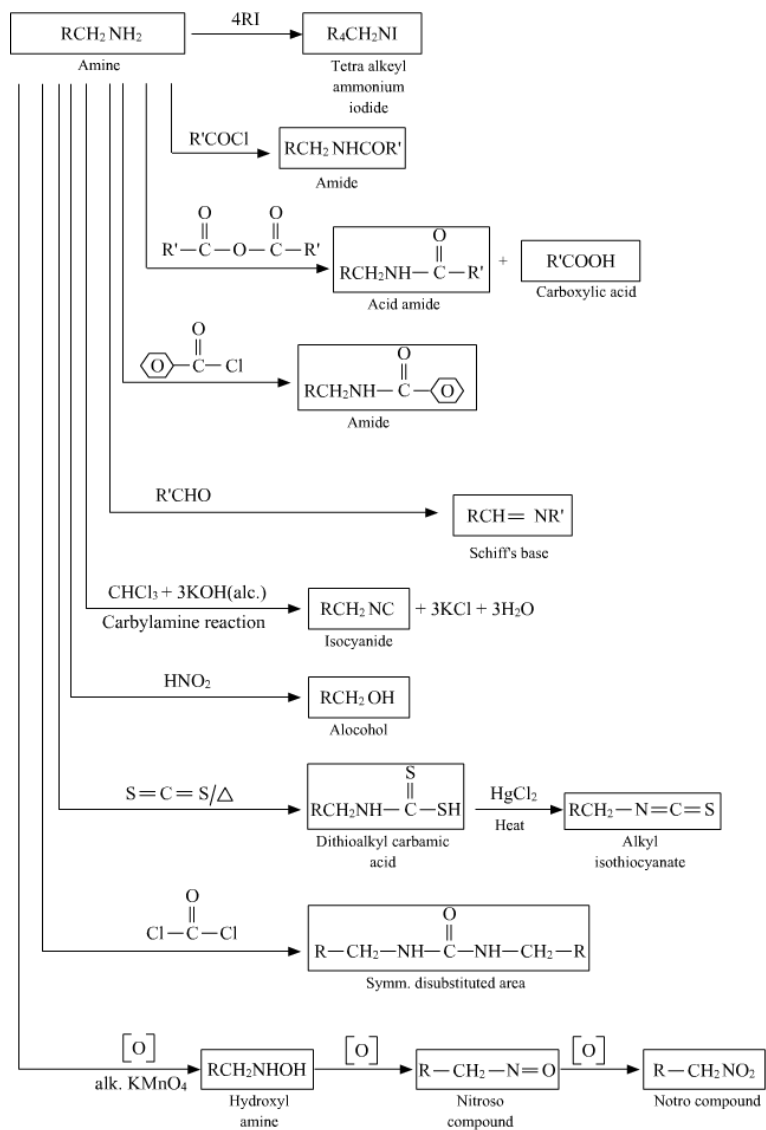
SCHEME – VI: Conversion related to aldehydes



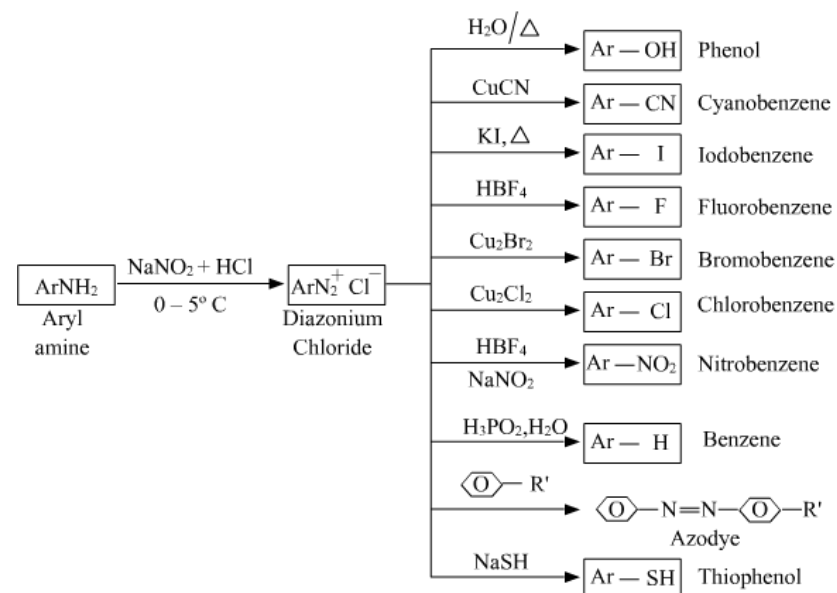
SCHEME – VI: Conversion related to carboxylic acids



SCHEME – VII: Conversion related to alkyl amines

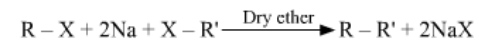
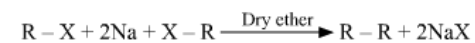


SCHEME – VIII: Conversion related to aryl amines

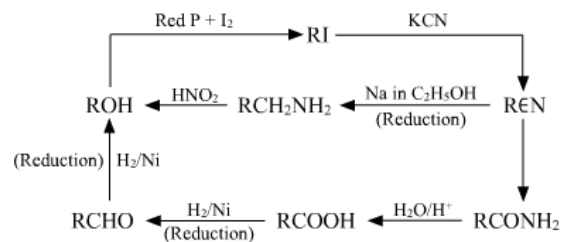


ASCENDING SERIES

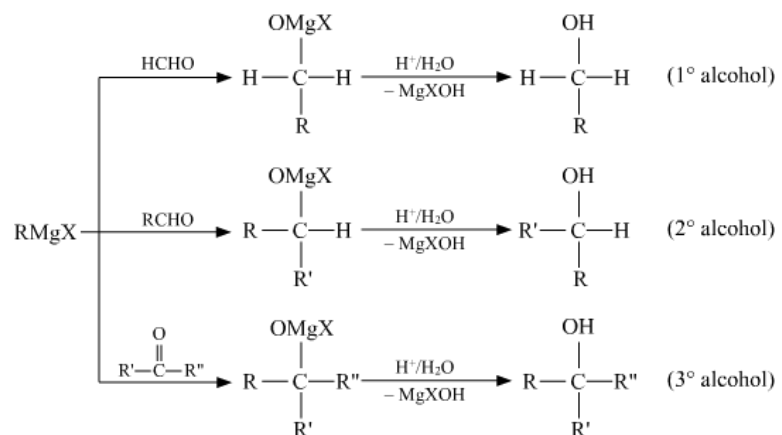
(1) By Wurtz Reaction



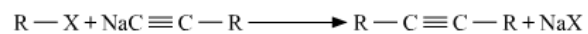
(2) By Using Cyanide



(3) By using Grignard's Reagent



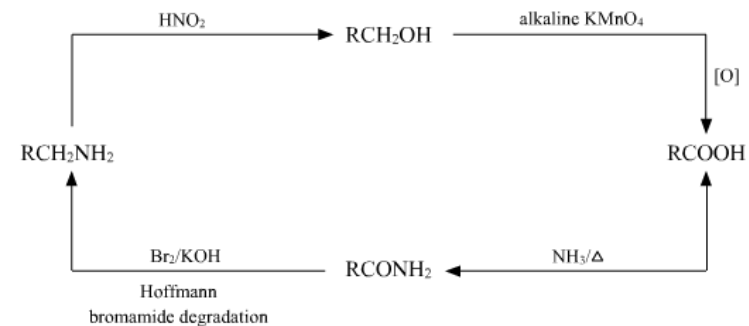
(4) By using Sodium Alkynides



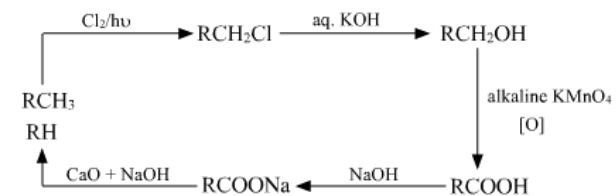
This reaction is used for terminal alkynes.

DESCENT OF SERIES

(1) Hoffmann Bromamide reaction



(2) Decarboxylation reaction



I. R – Cl vs R – Br vs R – I (R ≡ alkyl or aryl)

SNo.	Test	R – Cl	R – Br	R – I
a)	Dil AgNO ₃	$R-Cl \xrightarrow{AgNO_3} AgCl$ (White ppt)	$R-Br \xrightarrow{AgNO_3} AgBr$ (Pale yellow ppt)	$R-I \xrightarrow{AgNO_3} AgI$ (Yellow ppt)
b)	NH ₄ OH test	above ppt of AgCl + liq. NH ₃ or NH ₄ OH ↓ ppt dissolves	above ppt of AgBr + liq. NH ₃ or NH ₄ OH ↓ ppt partially dissolves	above ppt + liq. NH ₃ or NH ₄ OH ↓ ppt remains insoluble

II. Ethylidene chloride (Geminal) vs Ethylene Dichloride (Vicinal)

SNo.	Test	CH ₃ – CH $\begin{matrix} \diagup Cl \\ \diagdown Cl \end{matrix}$ (Ethylidene chloride)	Cl – CH ₂ – CH ₂ – Cl Ethylene dichloride
a)	Aq KOH test (Hydrolysis)	$CH_3 - CH \begin{matrix} \diagup Cl \\ \diagdown Cl \end{matrix} \xrightarrow{aq\ KOH} [CH_3 - CH \begin{matrix} \diagup OH \\ \diagdown OH \end{matrix}]$ $\xrightarrow{H_2O} CH_3 - C \begin{matrix} \diagup H \\ \diagdown O \end{matrix}$ $\xrightarrow{-H_2O} \text{2, 4 dinitro phenyl hydrazine} \rightarrow \text{Yellow ppt}$	$\begin{matrix} CH_2 - CH_2 \\ \quad \\ Cl \quad Cl \end{matrix} \xrightarrow{aq\ KOH} \begin{matrix} CH_2 - CH_2 \\ \quad \\ OH \quad OH \end{matrix}$ $\xrightarrow{\text{2, 4 dinitrophenyl hydrazine}} \text{No reaction}$

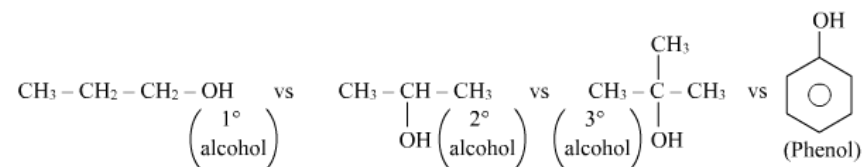
III. CHCl₃ vs CH₃Cl/CCl₄/CH₃OH

SNo.	Test	CHCl ₃	CH ₃ Cl/CCl ₄ /CH ₃ OH
a)	Carbylamine test	$R-NH_2 + 3KOH + CHCl_3 \xrightarrow{(aq)} R-NC + 3KCl + 3H_2O$ (1° amine) alkyl isocyanide Pungent Smelling	No reaction

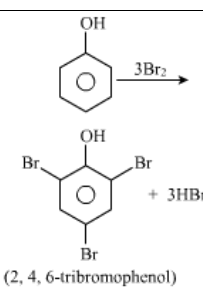
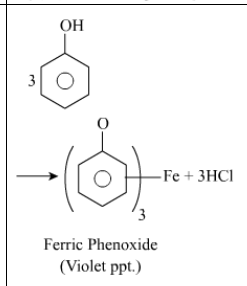
IV. CH₃ – CH₂ – OH (Alcohol) vs CH₃ – O – CH₃ (Ether)

SNo.	Test	CH ₃ – CH ₂ – OH	CH ₃ – O – CH ₃
a)	Na metal test	$CH_3 - CH_2 - OH + Na \xrightarrow{+ve} CH_3 - CH_2 - ONa + \frac{1}{2} H_2 \uparrow$	
b)	Iodoform test (for alcohols having CH ₃ – CH – OH)	$CH_3CH_2OH + 6NaOH + 4I_2 \xrightarrow{\Delta} CHI_3 \downarrow + HCOONa + 5NaI + 5H_2O$	

V.

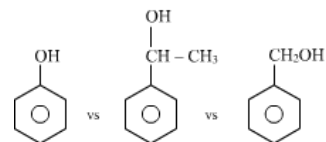


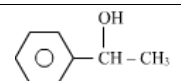
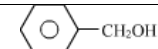
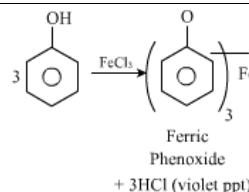
SNo.	Test	$CH_3 - CH_2 - \underset{\substack{ \\ OH}}{CH}$ (1° alcohol)	$CH_3 - \underset{\substack{ \\ OH}}{CH} - CH_3$ (2° alcohol)	$CH_3 - \underset{\substack{ \\ OH}}{\overset{\substack{CH_3 \\ }}{C}} - CH_3$ (3° alcohol)	Phenol
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a)	Lucas Test (Conc. HCl + anhyd ZnCl ₂)	Turbidity appears on heating	Turbidity appears within in 5 – 10 min.	Turbidity appears spontaneously	No appearance of turbidity (-ve)
b)	Iodoform test	(-ve)	$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\ \\ \text{OH} \end{array} + 6\text{NaOH} + 4\text{I}_2 \longrightarrow$ $\text{CHI}_3(\downarrow) + \text{HCOONa}$ Yellow + 5NaI + 5H ₂ O	(-ve)	(-ve)
c)	Br₂ water test	(-ve)	(-ve)	(-ve)	 (2, 4, 6-tribromophenol)
d)	Neutral FeCl₃ Test	(-ve)	(-ve)	(-ve)	 Ferric Phenoxide (Violet ppt.)
e)	Litmus Test	(-ve)	(-ve)	(-ve)	Turns blue litmus red (-ve) (-ve) (-ve) WEAK ACIDS

f)	Victor Meyer Test	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \\ \downarrow \text{P} + \text{I}_2 \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{I} \\ \downarrow -\text{AgI} \uparrow \text{AgNO}_2 \\ \text{CH}_3\text{CH}_2\text{CH}_2\text{NO}_2 \\ \downarrow \text{HNO}_2 \\ \text{CH}_3 - \text{C} = \text{NO}_2 \\ \\ \text{NOH} \\ \text{Nitrolic Acid} \\ \downarrow \text{NaOH} \\ \text{Blood Red Colouration} \end{array}$	$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\ \\ \text{OH} \\ \downarrow \text{P} + \text{I}_2 \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \\ \\ \text{I} \\ \downarrow -\text{AgI} \uparrow \text{AgNO}_2 \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \\ \\ \text{NO}_2 \\ \downarrow \text{HNO}_2 \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ \\ \text{N} = \text{O} \\ \text{Pseudonitrol} \\ \downarrow \text{NaOH} \\ \text{Blue Colouration} \end{array}$	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{OH} \\ \\ \text{CH}_3 \\ \downarrow \text{P} + \text{I}_2 \\ \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{I} \\ \\ \text{CH}_3 \\ \downarrow -\text{AgI} \uparrow \text{AgNO}_2 \\ \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{NO}_2 \\ \\ \text{CH}_3 \\ \downarrow \text{HNO}_2 \\ \text{No reaction} \\ \downarrow \text{NaOH} \\ \text{Colourless} \end{array}$	(-ve)
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VI.



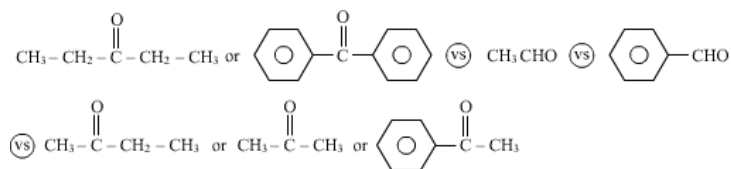
SNo.	Test		 (-ve)	 (-ve)
a)	Litmus Test	Turns blue Litmus to red	(-ve)	(-ve)
b)	Neutral FeCl₃ test	 Ferric Phenoxide + 3HCl (violet ppt)	(-ve)	(-ve)

c)	Iodoform Test	(-ve)	$\text{C}_6\text{H}_5\text{CH(OH)CH}_3 + 6\text{NaOH} + 4\text{I}_2 \longrightarrow \text{C}_6\text{H}_5\text{COONa} + \text{CHI}_3 \downarrow + 5\text{NaI} + 5\text{H}_2\text{O}$ (Yellow ppt)	(-ve)
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VII. HCHO vs CH₃CHO

SNo.	Test	HCHO	CH ₃ CHO
a)	Iodoform test	(-ve)	(+ve) $\text{CH}_3\text{CHO} + 4\text{NaOH} + 3\text{I}_2 \longrightarrow \text{CHI}_3 \downarrow + \text{HCOONa} + 3\text{NaI} + 3\text{H}_2\text{O}$ (Yellow)
b)	Liquor Ammonia Test	$6\text{HCHO} + 4\text{NH}_3 \longrightarrow (\text{CH}_2)_6\text{N}_4 + 6\text{H}_2\text{O}$ Hexamethylene tetramine (urotropine)	$\text{CH}_3\text{CHO} + \text{NH}_3 \longrightarrow \text{CH}_3\text{CH}=\text{NH}$ Addition product

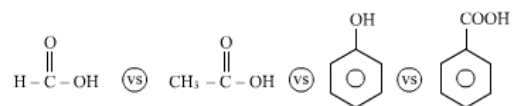
VIII.



SNo.	Test	$\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ or $\text{C}_6\text{H}_5\text{COCH}_2\text{CH}_3$	CH ₃ CHO		
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a)	Iodoform test	(-ve)	(+ve) $\text{CH}_3\text{CHO} + 4\text{NaOH} + 3\text{I}_2 \longrightarrow \text{CHI}_3 \downarrow + \text{HCOONa} + 3\text{NaI} + 3\text{H}_2\text{O}$ (Yellow)	(-ve)	(+ve) $\text{CH}_3\text{COCH}_2\text{CH}_3 + 6\text{NaOH} + 3\text{I}_2 \longrightarrow \text{CHI}_3 \downarrow + \text{CH}_3\text{CH}_2\text{COONa} + 3\text{NaI} + 3\text{H}_2\text{O}$ (Yellow)
b)	Tollen's reagent (amm. silver nitrate)	(-ve)	(+ve) $\text{CH}_3\text{CHO} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 2\text{OH}^- \longrightarrow \text{CH}_3\text{COO}^- + \text{NH}_4^+ + 2\text{Ag} \downarrow + \text{H}_2\text{O} + 3\text{NH}_3$	(+ve) $\text{C}_6\text{H}_5\text{CHO} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 2\text{OH}^- \longrightarrow \text{C}_6\text{H}_5\text{COO}^- + \text{NH}_4^+ + 2\text{Ag} \downarrow + \text{H}_2\text{O} + 3\text{NH}_3$	(-ve)
c)	Fehling's solution (copper sulphate + sodium potassium tartarate)	(-ve)	(+ve) $\text{CH}_3\text{CHO} + 2[\text{Cu}(\text{OH})_2] + \text{NaOH} \longrightarrow \text{CH}_3\text{COO}^- + \text{Na}^+ + \text{Cu}_2\text{O} \downarrow + 3\text{H}_2\text{O}$ (Red ppt)	(+ve) Oxidation is very difficult	(-ve)

IX.



SNo.	Test	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$		$\text{C}_6\text{H}_5\text{COOH}$
a)	Tollen's test	(+ve) $\text{HCOOH} + \text{Ag}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + 2\text{Ag}\downarrow$	(-ve)	(-ve)	(-ve)
b)	Fehling's Solution test	(+ve) $\text{HCOOH} + 2\text{CuO} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Cu}_2\text{O}\downarrow$ (Reddish Brown ppt)	(-ve)	(-ve)	(-ve)
c)	NaHCO₃ test	(+ve) $\text{HCOOH} + \text{NaHCO}_3 \rightarrow \text{HCOONa} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ (Brisk Effervescence)	(+ve) $\text{CH}_3\text{COOH} + \text{NaHCO}_3 \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ (Brisk Effervescence)	(-ve)	(+ve) $\text{C}_6\text{H}_5\text{COOH} + \text{NaHCO}_3 \rightarrow \text{C}_6\text{H}_5\text{COONa} + \text{H}_2\text{O} + \text{CO}_2\uparrow$ (Brisk Effervescence)

d)

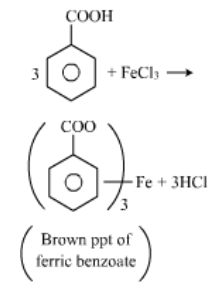
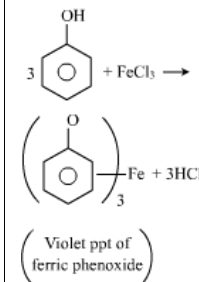
Neutral FeCl₃ test

(-ve)

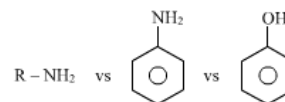
(-ve)

(+ve)

(+ve)



X.



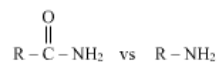
SNo.	Test	$\text{R}-\text{NH}_2$	$\text{C}_6\text{H}_5\text{NH}_2$	$\text{C}_6\text{H}_5\text{OH}$
a)	Bromine water	(-ve)	(+ve) $\text{C}_6\text{H}_5\text{NH}_2 + 3\text{Br}_2 \rightarrow \text{2,4,6-tribromo aniline (White ppt)} + 3\text{HBr}$	(+ve) $\text{C}_6\text{H}_5\text{OH} + 3\text{Br}_2 \rightarrow \text{2,4,6-tribromo phenol (White ppt)} + 3\text{HBr}$
b)	Neutral FeCl₃	(-ve)	(-ve)	(+ve)

				$3 \text{ C}_6\text{H}_5\text{OH} + \text{FeCl}_3 \rightarrow$ $\left(\text{C}_6\text{H}_5\text{O} \right)_3\text{Fe} + 3\text{HCl}$ (Violet ppt of ferric phenoxide)
c)	Carbylamine test	$\text{R}-\text{NH}_2 + 3\text{KOH} + \text{CHCl}_3 \xrightarrow{\text{(aq)}} \text{RNC} + 3\text{KCl} + 3\text{H}_2\text{O}$ (Pungent smelling)	$\text{C}_6\text{H}_5\text{NH}_2 + 3\text{KOH} + \text{CHCl}_3 \xrightarrow{\text{(aq)}} \text{C}_6\text{H}_5\text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$ Phenyl Isocyanide (Pungent smelling)	
d)	Azo Dye Test	Azo dye formed is unstable, so cannot be removed from solution	$\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[\text{HCl}]{\text{NaNO}_2} \text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow{0^\circ\text{C}} \text{C}_6\text{H}_5\text{N}=\text{N}-\text{C}_6\text{H}_4\text{OH}$ Azo dye	

XI. R – NH₂ vs R₂NH vs R₃N

SNo.	Test	R – NH ₂ (1° amine)	R ₂ NH (2° amine)	R ₃ N
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a)	Carbylamine Test	$\text{R}-\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\text{(aq)}} \text{R}-\text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$ alkyl isocyanide (Pungent smelling)	(-ve)	(-ve)
b)	Nitrous Acid Test	$\text{R}-\text{NH}_2 + \text{HO}-\text{N}=\text{O} \rightarrow \text{R}-\text{OH} + \text{N}_2\uparrow + \text{H}_2\text{O}$ Evolution of nitrogen	$\text{R}_2-\text{N}-\text{H} + \text{HO}-\text{N}=\text{O} \rightarrow \text{R}_2\text{N}-\text{N}=\text{O}$ N-nitroso dialkyl amine (Yellow oily liquid) + Phenol $\xrightarrow{\text{Warm}}$ Green colour	$\text{R}_3\text{N} + \text{HNO}_2 \xrightarrow{\text{Warm}} \text{R}_3\text{NHNO}_2$ (Water Soluble)
c)	Hinsberg's Test [Hinsberg's Reagent is a mixture of (i) Benzene sulphonyl chloride, (ii) KOH, and (iii) HCl]	$\text{R}-\text{NH}_2 + \text{C}_6\text{H}_5\text{SO}_2\text{Cl} \xrightarrow{-\text{HCl}} \text{R}-\text{N}(\text{H})-\text{SO}_2-\text{C}_6\text{H}_5$ N-alkylbenzene sulphonamide (Insoluble) $\xrightarrow[\text{KOH}]{-\text{H}_2\text{O}} \left[\text{R}-\text{N}(\text{SO}_2-\text{C}_6\text{H}_5)_2 \right] \text{K}^+$ Pot. Salt (Soluble in KOH) $\xrightarrow[\text{HCl}]{-\text{KCl}} \text{R}-\text{N}(\text{SO}_2-\text{C}_6\text{H}_5)_2$ N-alkylbenzene sulphonamide (insoluble)	$\text{R}_2-\text{NH} + \text{C}_6\text{H}_5\text{SO}_2\text{Cl} \xrightarrow{-\text{HCl}} \text{R}_2-\text{N}(\text{SO}_2-\text{C}_6\text{H}_5)$ N, N-dialkyl-benzene sulphonamide $\xrightarrow[\text{HCl}]{\text{KOH}} \text{No reaction (Insoluble)}$	$\text{R}_3-\text{N} + \text{C}_6\text{H}_5\text{SO}_2\text{Cl} \xrightarrow{\text{No reaction (Insoluble)}} \text{No reaction (Insoluble)}$ + HCl Trialkyl-ammonium chloride (Soluble in HCl)



SNo.	Test	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$	$\text{R}-\text{NH}_2$
a)	Litmus Test	No response to litmus	Red litmus changes to Blue
b)	Carbylamine test	(-ve)	$\text{R}-\text{NH}_2 + \text{CHCl}_3 + 3\text{KOH (aq)} \longrightarrow \text{RNC} + 3\text{KCl} + 3\text{H}_2\text{O}$ alkyl isocyanide Pungent Smelling

XIII. RNO_2 vs RONO

SNo.	Test	$\text{R}-\text{NO}_2$	RONO
a)	Reduction (Sn/HCl)	$\text{R}-\text{NO}_2 + 6\text{H} \longrightarrow \text{RNH}_2 + 2\text{H}_2\text{O}$	$\text{RONO} + 6\text{H} \longrightarrow \text{ROH} + \text{NH}_3 + \text{H}_2\text{O}$
b)	NaOH	Form's soluble sodium salt. $\text{R}-\text{N} \begin{matrix} \nearrow \text{O} \\ \searrow \text{O} \end{matrix} \xrightarrow{\text{NaOH}} \text{R}-\text{N} \begin{matrix} \nearrow \text{ONa} \\ \searrow \text{O} \end{matrix}$	Readily hydrolysed to give corresponding alcohol and sodium nitrite. $\text{RONO} + \text{NaOH} \longrightarrow \text{ROH} + \text{NaNO}_2$

XIV. RCN vs RNC

SNo.	Test	$\text{R}-\text{NO}_2$	RONO
a)	Solubility in water	Soluble	Insoluble
b)	Reduction followed by nitrous acid treatment	$\text{RCN} + 4\text{H} \longrightarrow \text{RCH}_2\text{NH}_2$ 1° amine $\downarrow \text{HNO}_2$ $\text{R}-\text{OH} + \text{N}_2\uparrow + \text{H}_2\text{O}$ Evolution of nitrogen	$\text{RCN} + 4\text{H} \longrightarrow \text{R}-\underset{\text{H}}{\text{N}}-\text{CH}_3$ $\downarrow \text{HNO}_2$ $\text{R}-\underset{\text{CH}_3}{\text{N}}-\text{N}=\text{O} + \text{H}_2\text{O}$ Yellow oily

c)	Hydrolysis	$\text{R}-\text{C}\equiv\text{N} \xrightarrow[\text{H}^+]{\text{H}_2\text{O}} \underset{\text{amide}}{\text{RCONH}_2}$ $\downarrow \text{H}^+/\text{H}_2\text{O}$ $\text{H}_3\text{N} + \text{RCOOH}$ Carboxylic acid	$\text{R}-\text{N}=\text{C} + 2\text{H}_2\text{O} \xrightarrow{\text{H}^+} \underset{\text{1° amine}}{\text{RNH}_2} + \underset{\text{Formic acid}}{\text{HCOOH}}$
d)	Heating	No effect	Alkyl cyanide is formed